INCH-POUND
A-A-58091C
7 AUG 2014
SUPERSEDING
A-A-58091B
26 JUN 2003

COMMERCIAL ITEM DESCRIPTION

TRAILER, TANK MOUNTED, RECOVERABLE AVIATION TURBINE FUEL (BOWSERS): 200, 400, AND 600 GALLON CAPACITY

The General Services Administration has authorized the use of this Commercial Item Description for all federal agencies.

1. SCOPE

1.1 Scope. This CID covers tanks mounted on trailers intended for receiving, vacuum pick-up, temporary storage, ground transport, and gravity draining of recoverable fuel: these will herein be referred to as the unit. These units are commonly referred to as bowsers. The units fulfill the requirements for standardized family of bowsers for general flight line and maintenance facility use. The units are intended for general flight line use in support of aircraft servicing and maintenance when temporary storage of fuel is requested, draining and depuddling aircraft fuel tanks in support of aircraft fuel cell and tank maintenance, removing fuel spills, pumping-out hydrant pits, and the like.

2 CLASSIFICATION.

2.1 Classification. The units are of the following three sizes:

2.1.1 Sizes. The sizes of the tanks are as follows:

Size I	200 gallon nominal capacity
Size II	400 gallon nominal capacity
Size III	600 gallon nominal capacity

Comments, suggestions, or questions on this document should be addressed to: AFLCMC/WNZEC, Robins AFB GA 31098-1813 or emailed to SPEC99@us.af.mil . Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil .

AMSC N/A FSC 2330

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:00Z
Check the source to verify that this is the current version before use.

3. SALIENT CHARACTERISTICS.

3.1 Description. The equipment shall be capable of operation within the accuracy's, limits, and specifications specified herein. The unit shall be designed and constructed for convenient, leak and weep free safe, effective reception and vacuum pickup, storage, ground transport and gravity drain of aviation turbine fuels. The storage tank on the unit shall be constructed with a double wall for secondary containment. The tank assembly shall have an operating pressure range of atmosphere (0 psig) to 8 inches Hg vacuum. The tank assembly shall be designed to withstand a pressure range of 2.5 psig to 12 inches Hg without leaking, weeping, fracturing, permanent deformation or evidence of impending failure. The unit shall withstand the strains, shocks and vibrations incidental to shipment, towing, fork lifting, and operation. All hose (air and fuel) connections shall be either male national pipe thread (NPT) or quick disconnect for air lines and cam locks or similar devices to allow for quick use on the flight line. Funnel lids and manway cover shall be provided with provisions for padlocking. If wheel chocks are implemented into the design, then drum type parking brakes (asbestos free), which are activated by a brake lever, shall be provided. The units shall meet the below listed minimum/maximum dimensions. The unit shall be designed to be corrosion resistant and shall be fabricated from 300 stainless steel. The exception to this shall be the telescoping funnel assembly which shall be fabricated from aluminum alloy. The undercarriage framework shall be fabricated from carbon steel. All components and materials (to include gaskets, fittings, valves, adhesives, etc.,) shall be proven fuel resistant materials.

3.2. Design and construction. The design shall promote cost effective, life-cycle sustainability by addressing considerations such as incorporating open standards and reducing waste, while satisfying system performance requirements. It shall be designed and constructed so that no parts will work loose in service, and to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. It shall be weatherproof and designed to prevent the intrusion of water, sand, and dust into critical operating components.

3.2.1 Fastening devices. All screws, bolts, nuts, pins, and other fastening devices shall be properly designed, manufactured, and installed with adequate means of preventing loss of torque or adjustment. Cotter pins, lock washers, or nylon patches shall not be used for this purpose, except for the attachment of trim items or as provided in commercial components. Tapped threads shall have a minimum thread engagement in accordance with Table I.

TABLE I. Minimum thread engagement.

Material	Minimum Thread Engagement
Steel	1.0 times the nominal fastener diameter
Cast iron, brass, or bronze	1.5 times the nominal fastener diameter
Aluminum, zinc, or plastic	2.0 times the nominal fastener diameter

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:00Z
Check the source to verify that this is the current version before use.

3.2.2 Welders and welding. All welders shall be certified to weld in accordance with AWS D1.1 and AWS D1.2, as applicable. The contractor shall make available to the Government certifications for all welders being utilized on the fuel bowser. Welding procedures and all welding on the fuel bowser shall be in accordance with AWS D1.1 and AWS D1.2, as applicable. The surface parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the welded parts. Welds shall transmit stress without cracking or permanent distortion when the parts connected by the welds are subjected to test, proof, and service loadings.

3.3 Components. The units shall consist essentially of the following major components listed in Table II.

Table II. Major components.

Component	**Paragraph Reference**
Trailer chassis	3.3.1
Tank assembly	3.3.2
Telescoping funnel assembly	3.3.3
Vacuum pump assembly	3.3.4
Vacuum pickup hose assembly	3.3.5
Aircraft interface vacuum assembly	3.3.6
Sampling device	3.3.7
Static grounding reels	3.3.8
Grounding receptacles	3.3.9

3.3.1 Trailer chassis. A trailer chassis shall be provided that shall securely mount all components specified herein. The trailer tow bar shall incorporate heavy duty pins (1 inch diameter or larger) and a lunette coupler.

3.3.2 Tank assembly. The unit shall be furnished with a rigid double walled tank. Tank nominal volume shall be as specified (see 2.1.1). Tank shall have a true volume not less than 110 percent of the tank's nominal volume. Space enclosed by tank sump and manway shall not be attributed to the true volume. The tank shall be circular, elliptical or semi-elliptical in cross section. The tank shall be baffled laterally.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:00Z
Check the source to verify that this is the current version before use.

3.3.3 Telescoping funnel assembly. A telescoping funnel assembly shall be provided mounted on the tank assembly of Sizes 400 and 600; Size 200 unit shall not have a tank assembly mounted telescoping funnel assembly. The funnel shall be a truncated cone not less than 15 inches in diameter and not less than 3 inches deep and shall have a bottom opening not less than 2-inches in diameter. Funnel assembly height shall be infinitely variable from 4 feet 2 inches to 16 feet measured from ground plane to the upper edge of the funnel rim.

3.3.4 Vacuum pump assembly. A vacuum pump assembly shall be provided. The vacuum pump assembly shall consist of a compressed air driven vacuum generator and a vacuum chamber. The vacuum chamber shall consist of the entire tank assembly, and be provided with four ports on the vacuum chamber that enable attaching drain hoses to the low point drains on the aircraft. The unit shall provide an assembly to collect foreign objects. The ports shall be attached to tubing that extends the height of the F.O.D. chamber and shall be designed to prevent splashing and free-fall of the fuel during draining into the tank assembly. Means shall be provided to interrupt the vacuum action should the tank assembly approach an over-fill condition; a manual reset shall be provided. The unit's vacuum pump shall produce no less than 12 inches of mercury (Hg) of lift and 80 standard cubic feet per minute (scfm) vacuum flow at the operational air input of 100 Pounds per Square Inch Gage (psig) and 60 scfm air consumption.

3.3.5 Vacuum pickup hose assembly. A vacuum pickup hose assembly shall be furnished. The hose assembly shall be not less than 50 feet long. The hose shall have a conductive/static dissipating cover. The vacuum pickup hose shall be adaptable to the unit's vacuum pump assembly on one end and no fittings on the opposite end.

3.3.6 Aircraft interface vacuum assembly. An aircraft interface vacuum assembly shall be furnished. The device shall interface to all aircraft tank drains while providing a leak-free connection and the ability to drain fuel with vacuum assist. The aircraft interface vacuum assembly shall be able to be attached to the unit at a distance no less than 35 feet.

3.3.7 Sampling device. A sampling device shall be provided and positioned to permit sampling of fuel as it flows out of the tank.

3.3.8 Static grounding reels. Each unit shall he provided with no less than two static grounding reels in accordance with A-A-50696. The reels shall have cables that are insulated with a corrosion resistant coating, reels shall be mounted vertically. Reels may be mounted inverted. Electrical resistance between reel base and the unit frame shall not exceed 0.5 ohms.

3.3.9 Grounding receptacles. Two grounding receptacles shall be mounted on the unit. Electrical resistance between the unit and the receptacle shall not exceed 0.5 ohms.

3.4 Materials, protective coatings, and finish.

3.4.1 Protective coatings. Materials that deteriorate when exposed to sunlight, weather, or operational conditions normally encountered during the service life of the item shall not be used or shall have means of protection against such deterioration that does not prevent compliance with the performance requirements specified herein. Protective coatings that chip, crack, or scale with age or extremes of climatic conditions or when exposed to heat shall not be used. Exposed

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:00Z
Check the source to verify that this is the current version before use.

surfaces of fasteners, handles, and fittings shall also be primed and painted. See 3.6.1 for prohibited hazardous materials.

3.4.2 Finish. The exterior finish color of the fuel bowser shall be grey, Color Number 26173 of FED-STD-595.

3.4.3 Exclusion of water. The design of the fuel bowser shall be such as to prevent water leaking into, or being driven into, any part of the fuel bowser interior when either in an operating or travelling configuration. All doors, panels, covers, etc., shall be provided with sealing arrangements such that the entry of water is minimized when these items are correctly closed. Particular care shall be taken to prevent wetting of heat and sound proofing materials. Sharp corners and recesses shall be avoided so that moisture and solid matter cannot accumulate to initiate localized attack. Sealed floors with suitable drainage shall be provided for storage compartments, and other areas in the fuel bowser that could collect and retain water.

3.4.4 Fluid traps and faying surfaces. There shall be no fluid traps on the fuel bowser. Faying surfaces of all structural joints, except welded joints, shall be sealed to preclude fluid intrusion.

3.4.5 Ventilation. Ventilation shall be sufficient to prevent moisture retention and buildup.

3.4.6 Drainage. Drain holes shall be provided to prevent collection or entrapment of water or other unwanted fluid in areas where exclusion is impractical. All designs shall include considerations for the prevention of water or fluid entrapment and ensure that drain holes are located to effect maximum drainage of accumulated fluids. The number and location of drain holes shall be sufficient to permit drainage of all fluids when the fuel bowser is on a 20 degree longitudinal slope facing both up and down and on an 11.5 degree side slope facing both directions. The minimum size of the drain holes shall be 0.25 inch.

3.5 Identification and information plates.

3.5.1 Markings. All external devices which require an operational or maintenance interface shall be marked in accordance with MIL-STD-130. Markings shall be applied with decals and shall be 1-inch high block letters unless prohibited by the available space. In such cases, the markings shall be the largest size possible. Markings, Information/Caution shall be Lusterless Black, Color Number 37038 of FED-STD-595, and Markings, Warning/Danger shall be Lusterless Red, Color Number 31136 of FED-STD-595. The center of gravity of the fuel bowser shall be stenciled on the unit within 1.0 inch of the calculated center of gravity.

3.5.2 Identification plate. An identification plate in accordance with MIL-STD-130 shall be securely attached to the fuel bowser in a readily accessible location. The identification plate shall contain the following information: Nomenclature, part number, serial number, date of manufacture, manufacturer's name, Commercial and Government Entity (CAGE) code, date of warranty expiration, and National Stock Number (NSN). The fuel bowser and any of its components for which the Government's unit cost is more than $5,000, is serially managed, or the procuring agency determines is mission essential, shall have Unique Identification (UID) (also known as Item Unique Identification (IUID)) information permanently affixed on or near

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:00Z
Check the source to verify that this is the current version before use.

the respective identification plate(s), marked in accordance with MIL-STD-130. UID information shall be included as both a bar code and human readable markings.

3.5.3 Transportation data plate. A transportation data plate shall be securely attached to the fuel bowser in a readily accessible location. The plate shall contain at least the following information:

a. Side and rear silhouette views of the fuel bowser.

b. Horizontal and vertical location of the center of gravity of the fuel bowser in air transportable configuration, marked on the silhouette views.

c. Shipping weight.

d. Loading cubage.

e. Overall height, width, and length.

f. Front and rear axle loads.

g. Tie down information.

3.6 Environment, Safety, and Occupational Health (ESOH).

3.6.1 Hazardous material. The design shall minimize and control hazards associated with the inclusion or use of hazardous or toxic materials and the generation of toxic or noxious gases. The fuel bowser shall not generate or use Class I or Class II Ozone Depleting Substances (ODS) during operation, maintenance, or disposal. The fuel bowser shall not contain or use hexavalent chromium. For the purposes of this requirement, the Class I ODS and controlled substances identified in Chapter 4 of AFI 32-7086 shall not be used in any system, component, or process.

3.6.2 Component protection. All space in which work is performed during operation, service, and maintenance shall be free of hazardous protrusions, sharp edges, or other features which may cause injury to personnel. All rotating and reciprocating parts and all parts subject to high operational temperatures or subject to being electrically energized, that are of such nature or so located as to be hazardous to personnel, shall be guarded or insulated to eliminate the hazard. All wires, cables, tubes, and hoses shall be supported and protected to minimize chafing and abrasion and shall be located so as to provide adequate clearance from moving parts and high operational temperatures. Grommets shall be provided wherever wires, cables, tubes, or hoses pass through bulkheads, partitions, or structural members.

3.6.3 Foreign object damage (FOD). All loose metal parts, such as pins or connector covers, shall be securely attached to the fuel bowser with wire ropes or chains. "Dog tag" style beaded chains shall not be provided. Removable panels, if provided, shall be attached with captive fasteners. Tire valve stem caps shall be made of plastic.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:00Z
Check the source to verify that this is the current version before use.

3.6.4 Human systems integration. The fuel bowser shall be designed in accordance with MIL-STD-1472 for ease of operation, inspection, and maintenance, including the use of arctic mittens and Mission-Oriented Protective Posture (MOPP) Level 4 Chemical Warfare Gear. Chemical Warfare Gear is not required for preventive maintenance or major corrective maintenance.

3.7 Environmental conditions.

3.7.1 Operating temperature range. The fuel bowser shall be capable of operating in ambient temperatures ranging from -25° F to 125° F.

3.7.2 Storage temperature range. The fuel bowser shall be capable of being stored in ambient temperatures ranging from -40° F to 160° F.

3.7.3 Precipitation.

3.7.3.1 Rain. The fuel bowser shall be capable of storage and operation during rainfall of 5-inches per hour for three consecutive hours and 10-inches per hour for 10 consecutive minutes, with winds of up to 35 knots; and with 6-inches of rain per hour impinging on the fuel bowser at angles from vertical to 45° for 30 consecutive minutes.

3.7.3.2 Snow. The fuel bowser shall be capable of storage and operation during accretion of wet snow up to 2-inches per hour for at least 12 hours.

3.7.3.3 Ice. The fuel bowser shall be capable of storage and operation with ice accretion up to 1.5-inches on exposed horizontal surfaces. An operator may use an ice scraper for five minutes during the start-up process.

3.7.4 Solar radiation. The fuel bowser shall not be adversely affected by full time exposure to solar radiation, such as those conditions encountered in desert environments.

3.7.5 Fungus. All materials used in the fuel bowser shall be fungus resistant or shall be suitably treated to resist fungus. Materials treated for fungus resistance shall retain their original electronic and physical properties, shall not present toxic hazards, and treatment shall last for the entire service life of the part. The fuel bowser shall be suitable for operation and storage in conditions encountered in a tropical environment.

3.7.6 Salt fog. The fuel bowser shall be capable of storage and operation in high temperature, high humidity, salt laden, sea coast environments without damage or deterioration of performance.

3.7.7 Sand and dust. The fuel bowser shall be capable of storage and operation during exposure to wind-blown sand or dust without damage or deterioration of performance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:00Z
Check the source to verify that this is the current version before use.

3.8 Weight and dimensions. Overall weight and dimensions in air transport configuration (see 3.9.2) shall fall within the following range listed in Table III.

Table III. Weight and dimensions.

SIZE	WEIGHT	DIMENSIONS
200	Length (Tow Bar Extended) Width Height Max Weight (empty)	140/160 inches 70/80 inches 50/80 inches 1500 lbs.
400	Length (Tow Bar Extended) Width Height Max Weight (empty)	140/180 inches 70/95 inches 50/80 inches 1800 lbs.
600	Length (Tow Bar Extended) Width Height Max Weight (empty)	140/200 inches 70/95 inches 50/80 inches 2300 lbs.

3.9 Transportability.

3.9.1 Surface transportability. The fuel bowser shall be transportable via all modes of surface shipment (highway, rail, and water) in accordance with MIL-STD-1366, and shall be capable of withstanding the mechanical shock and vibration characteristics of highway, rail, and water transport, except that design for rail impact testing (see 5.2.5 of MIL-STD-1366) is not required.

3.9.2 Air transportability. The fuel bowser shall be transportable on C-130, C-17, and C-5 aircraft in accordance with MIL-STD-1791. The tow bar shall be transported in its upright storage position. In all air transport configurations, the fuel bowser shall be capable of being restrained and withstanding, without loss of serviceability, 2.0 G up and 4.5 G down accelerations, and shall be capable of being restrained and withstanding, without loss of structural integrity, 3.0 G forward, 1.5 G aft, and 1.5 G lateral accelerations. The fuel bowser shall be equipped with pressure relief devices or configured for air transport to prevent any part from becoming a projectile in the event of catastrophic loss of aircraft cabin pressure. The fuel bowser shall roll on and off the aircraft, negotiating the required maximum ramp angles without shoring.

3.9.2.1 Shoring. The fuel bowser shall be air transportable without shoring.

3.9.2.3 Tire pressure. Tire pressure shall not exceed 100 pounds per square inch (psi) and shall not be reduced for air transport.

3.9.2.4 Winch points. Winch points shall be provided at the front and rear of the fuel bowser for aircraft loading using the aircraft's winch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:00Z
Check the source to verify that this is the current version before use.

3.9.2.5 Equipment removal and reconfiguration. Preparation for air transport shall take no more than one man-hour and restoration to operating configuration shall take no more than one hour for one person using common hand tools (see 7.3.1). All equipment removed shall be stored on the fuel bowser; caps and plugs shall permit moving and storage in transport configuration.

3.9.3 Tie downs. The fuel bowser shall be symmetrically restrained during air and ground transport. Tie down points shall be rated at a minimum of 10,000 pounds, marked for capacity, with a clear opening compatible with MIL-DTL-25959 and MIL-PRF-27260 tie down devices. Each end of each tie down device shall terminate at a tie down point and not pass through any other tie down point. There shall be no interference between tie down devices and the fuel bowser. The tie down provisions shall be in accordance with 4.1 through 4.12 of MIL-STD-209.

3.9.4 Lifting provisions. The fuel bowser shall be equipped with sufficient attachment points so located that it can be lifted by crane; each attachment point shall be marked "Lift Point". The lifting provisions shall be in accordance with 5.1 through 5.1.4 of MIL-STD-209.

3.10 Maintainability. The fuel bowser shall be designed for maintainability in accordance with 5.9 through 5.9.18 of MIL-STD-1472; forces shall not exceed those specified for both males and females.

3.10.1 Inspection and servicing provisions.

a. Pre-use inspections and servicing tasks shall not require tools.

b. Routine service and preventive maintenance shall not require special tools (see 7.3.2).

c. Drain plugs and filters shall be directly accessible from the ground and oriented to have unimpeded drainage to a catch pan.

d. The fuel bowser shall be designed with maximum usage of sealed lifetime lubrication bearings.

3.10.2 Special tools. The design of the item shall minimize the requirement for special tools (see 7.3.2). All special tools shall be provided with, and stored on, the fuel bowser.

3.11 Mobility/ Stability The fuel bowser shall be in accordance with SAE AS8090 for Type II, Group C mobility except as otherwise specified herein. The unit shall be designed and constructed as a full trailer, with wheels and supported by its own suspension system.

3.11.1 Terrain.

3.11.1.1 Paved surfaces. The fuel bowser shall be towable at speeds up to 20.0 miles per hour (mph) over paved surfaces.

3.11.1.2 Graded gravel. The fuel bowser shall be towable at speeds up to 10.0 mph over graded gravel.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:00Z
Check the source to verify that this is the current version before use.

3.11.1.3 Belgian block. The fuel bowser shall be towable at speeds up to 8.0 mph over Belgian block.

3.11.2 Towing force. The towing force shall be in accordance with 3.3.4 of SAE AS8090.

3.11.3 Clearances. Ground clearance, ramp breakover angle, and angles of approach and departure shall be in accordance with 3.3.9 of SAE AS8090.

3.11.4 Suspension. The fuel bowser shall have a suspension system in accordance with 3.4.1.2 of SAE AS8090.

3.11.5 Brakes.

3.11.5.1 Parking brakes. The fuel bowser shall have parking brakes in accordance with 3.6.1 of SAE AS8090.

3.11.5.2 Service brakes. The fuel bowser shall have service brakes in accordance with 3.6.2 of SAE AS8090.

3.11.6 Bumper. The fuel bowser shall have a rear bumper in accordance with 3.7 of SAE AS8090.

3.11.7 Fenders. The fuel bowser shall have fenders in accordance with 3.9 of SAE AS8090.

3.11.8 Mudflaps. The fuel bowser shall have mudflaps in accordance with 3.12 of SAE AS8090.

3.11.9 Pintle hook. The rear pintle hook, specified in 3.13 of SAE AS8090, shall not be provided; the fuel bowser shall be towed singly, not in trains, as specified in 3.3.8 of SAE AS8090.

3.11.10 Towbar. The fuel bowser shall be equipped with a towbar in accordance with 3.17.1 of SAE AS8090. It shall have a lunette eye in accordance with A-A-52464 and accommodate any pintle hook height from 22 through 30 inches. When in operation or storage (that is, not being towed), the towbar shall be stowed upright, secured, and not hinder visibility, operation, or maintenance.

3.11.11 Tires and wheels. The fuel bowser shall be equipped with single or dual tubeless steel belted radial tires with on/off-road type tread mounted on steel disc wheel assemblies. Tire and wheel assemblies shall be identical at all wheel positions. Tires and wheels shall be in accordance with the *Tire and Rim Association Year Book* requirements for this application.

3.12 Workmanship. The fuel bowser, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Workmanship objectives shall include freedom from blemishes, defects, burrs and sharp corners and edges; accuracy of dimensions, surface finish, and radii of fillets; thoroughness of welding, painting, and riveting; marking of parts and assemblies; and proper alignment of parts and tightness of assembly fasteners.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:00Z
Check the source to verify that this is the current version before use.

3.12.1 Bolted connections. Bolt holes shall be accurately punched or drilled and shall be deburred. Threaded fasteners shall be tight and shall not work loose during testing or service usage.

3.12.2 Riveted connections. Rivet holes shall be accurately punched or drilled and shall be deburred. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the component.

3.12.3 Gear and lever assemblies. Gear and lever assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots, or other irregularities. Where required for accurate adjustment, gear assemblies shall be free of excessive backlash.

3.12.4 Cleaning. The fuel bowser shall be thoroughly cleaned. Loose, spattered, or excess solder; welding slag; stray bolts, nuts, and washers; rust; metal particles; pipe compound; and other foreign matter shall be removed during and after final assembly.

4. REGULATORY REQUIREMENTS.

4.1 Recycled recovered materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with 23.403 of the Federal Acquisition Regulation (FAR). However, used, rebuilt, or refurbished items shall not be provided.

4.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100% of EPA designated product purchase that are included in the Comprehensive Procurement Guidelines list that contains recovered materials, unless the item cannot be acquired: a) competitively within a reasonable timeframe; b) meet appropriate performance standards, or c) at a reasonable price. The prime contractor is responsible for ensuring that all subcontractors comply with this requirement.

5. PRODUCT CONFORMANCE PROVISIONS.

The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer’s own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace, modified as necessary to comply with the requirements herein. The Government reserves the right to require proof of such conformance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:00Z
Check the source to verify that this is the current version before use.

5.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First production inspection (see 5.2).

b. Conformance inspection (see 5.3).

5.2 First production inspection. The first production fuel bowsers shall be subjected to the analyses, demonstrations, examinations, and tests described in 5.5.1 through 5.5.10.8. The contractor shall provide or arrange for all test equipment and facilities.

5.3 Conformance inspection. Each production fuel bowser shall be subjected to the examination described in 5.5.1.

5.4 Inspection requirements.

5.4.1 General inspection requirements. Apparatus used in conjunction with the inspections specified herein shall be laboratory precision type, calibrated at proper intervals to ensure laboratory accuracy.

5.4.2 Data. During all testing specified herein, at least the following data, unless not applicable, shall be recorded at intervals not to exceed 30 minutes. Additional data or shorter intervals shall be provided as appropriate for any specific test.

a. Date.

b. Time started.

c. Time finished.

d. Ambient temperature.

e. Ambient humidity.

5.4.3 Test rejection criteria. Throughout all tests specified herein, the fuel bowser shall be closely observed for the following conditions, which shall be cause for rejection.

a. Failure to conform to design or performance requirements specified herein or in the contractor's technical proposal.

b. Any spillage or leakage of any liquid, including fuel, coolant, lubricant, or hydraulic fluid, under any condition, except as allowed herein.

c. Structural failure of any component, including permanent deformation, or evidence of impending failure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:00Z
Check the source to verify that this is the current version before use.

d. Evidence of excessive wear. If excessive wear is suspected, the original equipment manufacturer's (OEM's) specifications or tolerances shall be utilized for making a determination.

e. Evidence of corrosion or deterioration.

f. Misalignment of components.

g. Conditions that present a safety hazard to personnel during operation, servicing, or maintenance.

h. Interference between the fuel bowser components or between the fuel bowser, the ground, and all required obstacles, with the exception of normal contact by the tires.

i. Evidence of undesirable mobility characteristics, including instability in handling during cornering, braking, and while traversing all required terrain.

5.5 Test methods.

5.5.1 Examination of product. Each fuel bowser shall be examined to verify compliance with the requirements herein prior to accomplishing any other demonstrations or tests listed in 5.5. A contractor-generated, Government-approved checklist (part of the test procedure) shall be used to identify each requirement not verified by an analysis, certification, demonstration, or test, and shall be used to document the examination results. Particular attention shall be given to materials, workmanship, dimensions, surface finishes, protective coatings and sealants and their application, welding, fastening, and markings. Proper operation of each fuel bowser function shall be verified. Certifications and analyses shall be provided in accordance with Table IV. Each production fuel bowser shall be inspected to a Government-approved reduced version of the checklist.

TABLE IV. Certifications and analyses.

Paragraph	Required Certifications and Analyses
3.7.3.2 Snow.	Contractor analysis of the snow load requirement (see 5.5.3.3.2).
3.7.4 Solar radiation.	Contractor certification that the fuel bowser performance is not adversely affected by full time exposure to solar radiation, such as those conditions encountered in desert environments.
3.7.5 Fungus.	Contractor certification that the materials used in construction of the fuel bowser are fungus resistant or suitably treated to resist fungus.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:00Z
Check the source to verify that this is the current version before use.

TABLE IV. Certifications and analyses – Continued.

Paragraph	Required Certifications and Analyses
3.9.1 Surface transportability.	Contractor surface transportability analysis (see 5.5.5.1.1) and certification that the fuel bowser is transportable via all modes of surface shipment (highway, rail, and water) in accordance with MIL-STD-1366, and shall be capable of withstanding the mechanical shock and vibration characteristics of highway, rail, and water transport.
3.9.2 Air transportability. 3.9.2.1 Shoring., and 3.9.2.4 Winch points.	Contractor air transportability analysis (see 5.5.5.2).
3.9.3 Tie downs.	Contractor tie down provision analysis (see 5.5.5.4.1).
3.9.4 Lifting provisions.	Contractor lifting provision analysis (see 5.5.5.5).

5.5.2 Noise test. The unit under any operating condition shall not exceed 84 decibels dBA slow scale measured not more than four feet from the perimeter of the unit with tow bar raised and at between 4 and 5 feet ground level.

5.5.3 Environmental testing.

5.5.3.1 High temperature storage and operation test. A first production fuel bowser shall be tested in accordance with MIL-STD-810, Method 501.6, Procedures I and II, to demonstrate compliance with the high temperature storage and operating requirements of 3.7.1 and 3.7.2. Test duration shall be one 24-hour cycle for each procedure beginning no less than two hours after test item temperature stabilization.

5.5.3.2 Low temperature storage and operation test. A first production fuel bowser shall be tested in accordance with MIL-STD-810, Method 502.6, Procedures I and II, to demonstrate compliance with the low temperature storage and operating requirements of 3.7.1 and 3.7.2. Test duration shall be one 24-hour cycle for each procedure beginning no less than two hours after test item temperature stabilization.

5.5.3.3 Precipitation.

5.5.3.3.1 Rain test. A first production fuel bowser shall be tested in accordance with MIL-STD-810, Method 506.6, Procedure I, to demonstrate compliance with 3.7.3.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:00Z
Check the source to verify that this is the current version before use.

5.5.3.3.2 Snow load analysis. An engineering analysis shall be performed to demonstrate compliance with the snow load requirement of 3.7.3.2, using a specific gravity of snow of 0.1 (Ref. 5.3 of MIL-STD-810, Part Three).

5.5.3.3.3 Ice accretion test. A first production fuel bowser shall be tested in accordance with MIL-STD-810, Method 521.4 with an ice thickness of 1.5-inches to demonstrate compliance with the ice accretion requirement of 3.7.3.3. The contractor shall identify those areas of the fuel bowser where ice removal is required prior to operation.

5.5.3.4 Salt fog test. A first production fuel bowser shall be tested in accordance with MIL-STD-810, Method 509.6, to demonstrate compliance with 3.7.6. Test duration shall be alternating 24-hour periods of salt fog exposure and drying conditions for 24-hour periods (two wet and two dry).

5.5.3.5 Sand and dust test. A first production fuel bowser shall be tested in accordance with MIL-STD-810, Method 510.6, Procedures I (12 hours) and II (90 minutes per side), to demonstrate compliance with 3.7.7.

5.5.4 Weight and dimension tests.

5.5.4.1 Weight and center of gravity test. The weight, center of gravity, and axle weights of a first production fuel bowser shall be measured to demonstrate compliance with the weight requirement of 3.8 and the axle weight requirement of 3.5.2.2.

5.5.4.2 Dimension measurement. A first production fuel bowser shall be measured to demonstrate compliance with the dimensional requirements of 3.8.

5.5.5 Transportability verification.

5.5.5.1 Surface transportability verification.

5.5.5.1.1 Surface transportability analysis. An engineering analysis shall be performed to demonstrate compliance with 3.9.1. The engineering analysis shall utilize the data for road transportation in accordance with MIL-STD-810.

5.5.5.2 Air transportability analysis. An engineering analysis shall be performed to demonstrate compliance with the air transportability requirements of 3.9.2, 3.9.2.1, and 3.9.2.4. The analysis shall include the tie downs and all major components and their ability to withstand the accelerations specified in 3.9.2. The analysis shall demonstrate the accessibility and structural adequacy of the winch points specified in 3.9.2.4. The evaluation shall also include a dimensional analysis for the fuel bowser while traversing the ramp and while loaded aboard C-130, C-17, and C-5 aircraft.

5.5.5.3 Equipment removal and reconfiguration demonstration. A first production fuel bowser shall be configured for transport on C-130, C-17, and C-5 aircraft and then reconfigured for operation to demonstrate compliance with 3.9.2.5. It shall be demonstrated that the forces required do not exceed those allowed in MIL-STD-1472.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:00Z
Check the source to verify that this is the current version before use.

5.5.5.4 Tie down provision verification.

5.5.5.4.1 Tie down provision analysis. An engineering analysis shall be performed to demonstrate compliance with the tie down provision requirements of 3.9.3.

5.5.5.4.2 Tie down provision test. A first production fuel bowser shall be tested to demonstrate compliance with the tie down provision requirements of 3.9.3.

5.5.5.5 Lifting provision verification.

5.5.5.5.1 Lifting provision analysis. An engineering analysis shall be performed to demonstrate compliance with the lifting provision requirements of 3.9.4.

5.5.5.5.2 Lifting provision test. A first production fuel bowser shall be tested to demonstrate compliance with the lifting provision requirements of 3.9.4.

5.5.6 Maintainability demonstration. All recommended preventive maintenance tasks shall be performed and the task times shall be recorded. The recommended frequencies of the preventive maintenance tasks and the times recorded to accomplish the tasks shall be used to develop an expected value of preventive maintenance time per measure of use, such as calendar time or hours of operation. It shall be demonstrated that the forces required do not exceed those allowed in MIL-STD-1472. All preventive maintenance tasks recommended to be performed daily and at the routine PMI shall also be performed by personnel wearing arctic mittens.

5.5.7 Mobility tests. A first production fuel bowser shall be tested in accordance with Table VIII of SAE AS8090 to demonstrate compliance with the mobility requirements of 3.9—3.9.2, 3.9.4, and 3.9.5. Note that the maximum towing force is 75 pounds per ton of gross weight, in accordance with 3.3.4 of SAE AS8090, rather than 50 pounds per ton of gross weight as stated in Table VIII of SAE AS8090.

5.5.8 Brake tests.

5.5.8.1 Parking brake test. The fuel bowser shall be tested in accordance with 4.5.7.1 of SAE AS8090 to demonstrate compliance with 3.11.6.1.

5.5.8.2 Service brake test. The fuel bowser shall be tested in accordance with 4.5.7.2 and 4.5.8 of SAE AS8090 to demonstrate compliance with 3.11.6.2.

5.5.9 Test fluid. The test fluid shall be water.

5.5.10 Test methods. The following paragraph ordering does not establish sequence of testing.

5.5.10.1 Examination of unit. The unit shall be tested to demonstrate that the tank volume to confirm adherence with specified requirements not otherwise tested. This examination shall be accomplished using a contractor developed checklist that shall identify specific unit attributes for which examination is made and results obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:00Z
Check the source to verify that this is the current version before use.

5.5.10.2 Pressure test. The tank assembly, with vents and valves closed; and test plates used in place of manway covers and the like, shall be pressure tested at no less than 2.5 psig for 30 minutes and vacuum tested at 12 inches Hg for 30 minutes without leakage, permanent deformation, malfunction or evidence of impending failure. The supply air system shall be pressure tested at two times the operating pressure for 30 minutes without leakage, permanent deformation, malfunction or evidence of impending failure.

5.5.10.3 Vacuum test. The pumping assembly vacuum chamber with valves closed shall be subjected to an internal vacuum of 12 inches Hg. Once this vacuum is attained and the vacuum source valve is off, shall be maintained for 30 minutes. The initial and final vacuum shall be recorded. There shall be no air leakage, any permanent deformation or evidence of impending failure to the unit

5.5.10.4 Continuity test. Electrical continuity across all component connections shall be demonstrated.

5.5.10.5 Tank volume test. The unit shall be tested to demonstrate that the tank volume is not less than 110 percent of rated (nominal) volume. Calculations satisfactory to the procuring activity may be substituted for this test.

5.5.10.6 Noise test. The unit shall be tested to demonstrate that the noise level meets the noise limit specified by paragraph 3.10 herein.

5.5.10.7 Stability test. The unit shall be tested to demonstrate that the unit meets the stability requirements as stated in paragraph 3.11 herein.

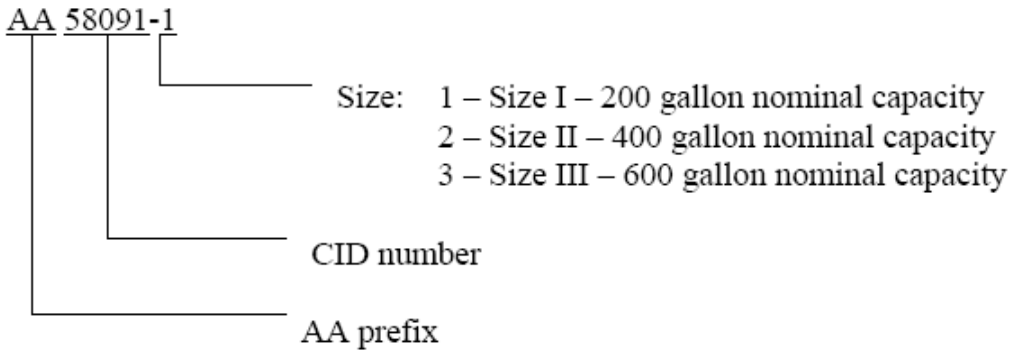
5.5.10.8 Funnel static load tests. The funnel assemblies shall be tested to demonstrate compliance with the static load requirements as stated in paragraph 3.3.3 herein.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:00Z
Check the source to verify that this is the current version before use.

7. NOTES.

7.1 Part or Identifying Number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor. The PIN to be used for trailer mounted tanks acquired to this CID is created as follows:



7.1.1 National Stock Number NSN. The following is a list of NSNs assigned that correspond to this CID. The list may not be indicative of all possible NSNs associated with this CID.

NSN	2330-01-446-5314
NSN	2330-01-245-9458

7.2 Source of documents.

7.2.1 Department of Defense and Federal documents. Department of Defense and Federal documents are available online at https://assist.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia PA 19111-5094.

7.2.2 AFI. Copies of AFIs are available online at http://www.e-publishing.af.mil/.

7.2.3 FAR. The FAR may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh PA 15250-7954. Electronic copies of the FAR may be obtained from https://www.acquisition.gov/far/.

7.2.4 AWS documents. Application for copies should be addressed to American Welding Society, 550 N.W. LeJeune Road, Miami FL 33126 or online at http://www.anweld.org/ .

7.2.5 SAE documents. Application for copies should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale PA 15096 or online at http://www.sae.org/ .

7.2.6 TRA Standards. Copies of TRA standards may be obtained from www.us-tra.org/ or the Tire and Rim Association, 175 Montrose West, Suite 150, Copley, OH 44321.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:00Z
Check the source to verify that this is the current version before use.

7.3 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this CID.

b. Finish color required (Color Number 26173 of FED-STD-595.) (see 3.2.1.3).

7.4 Definitions.

7.4.1 Common hand tool. A non-powered tool that is likely to be found in a typical mechanic's toolbox. Common hand tools include open end, boxed end, combination, socket (both 6- and 12-point in both standard and deep-well), and hex key wrenches, in SAE sizes up to and including 1-inch and metric sizes up to and including 25-mm; ratchet handles, extensions, and swivels; slotted and Phillips-head screwdrivers; regular and snap-ring pliers; and a ball-peen hammer.

7.4.2 Special tool. A tool that is not commercially and readily available from a source other than the fuel bowser contractor.

7.5 Key words.

Aviation Fuel
Bowser, Fuel
Mobile, Fuel
Tank, Fuel
Trailer, Fuel

7.6 Changes from previous issue. Marginal notations are not used in this in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:00Z
Check the source to verify that this is the current version before use.

Custodians:
Air Force - 184

Preparing Activity:
Air Force - 184

Reviewer:
Air Force - 99

Agent:
Air Force – 99
(Project 2330-2015-003)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/ .

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:00Z
Check the source to verify that this is the current version before use.